



Swathi Sri R 250701769-2025-CSE ▾

S2**Started on** Friday, 11 September 2026, 3:36 PM**State** Finished**Completed on** Friday, 11 September 2026, 3:37 PM**Time taken** 37 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Given two strings find the length of the common longest subsequence(need not be contiguous) between the two.

Example:

s1: ggtabe

s2: tgatasb

s1		a	g	g	t	a	b	
s2		g	x	t	x	a	y	b

The length is 4

Solveing it using Dynamic Programming

For example:

Input	Result
aab	2
azb	

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2  #include <string.h>
3
4  int max(int a, int b) {
5      return (a > b) ? a : b;
6  }
7
8  int main() {
9      char s1[1005], s2[1005];
10     if (scanf("%s %s", s1, s2) != 2) return 0;
11
12     int m = strlen(s1);
13     int n = strlen(s2);
14     int dp[n + 1];
15
16     for (int j = 0; j <= n; j++) {
17         dp[j] = 0;
18     }
19
20     for (int i = 1; i <= m; i++) {
21         int prev = 0;
22         for (int j = 1; j <= n; j++) {
23             int temp = dp[j];
24             if (s1[i - 1] == s2[j - 1]) {
25                 dp[j] = prev + 1;
26             } else {
27                 dp[j] = max(dp[j], dp[j - 1]);
28             }
29             prev = temp;
30         }
31     }
32
33     printf("%d\n", dp[n]);
34     return 0;
35 }
36

```

	Input	Expected	Got	
✓	aab azb	2	2	✓
✓	ABCD ABCD	4	4	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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